

Optimization Techniques in Municipal Solid Waste Management: A Systematic Review

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Abstract: As a consequence of human activity, waste generation is unavoidable, and its volume and complexity escalate with urbanization, economic progress, and the elevation of living standards in cities. Annually, the world produces about 2.01 billion tons of municipal solid waste, which often lacks environmentally safe management. The importance of solid waste management lies in its role in sustainable development, aimed at reducing the environmental harms from waste creation and disposal. With the expansion of urban populations, waste management systems grow increasingly complex, necessitating more sophisticated optimization strategies. This analysis thoroughly examines the optimization techniques used in solid waste management, assessing their application, benefits, and limitations by using PRISMA 2020. This study, reviewing the literature from 2010 to 2023, divides these techniques into three key areas: waste collection and transportation, waste treatment and disposal, and resource recovery, using tools like mathematical modeling, simulation, and artificial intelligence. It evaluates these strategies against criteria such as cost-efficiency, environmental footprint, energy usage, and social acceptability. Significant progress has been noted in optimizing waste collection and transportation through innovations in routing, bin placement, and the scheduling of vehicles. The paper also explores advancements in waste treatment and disposal, like selecting landfill sites and converting waste to energy, alongside newer methods for resource recovery, including sorting and recycling materials. In conclusion, this review identifies research gaps and suggests directions for future optimization efforts in solid waste management, emphasizing the need for cross-disciplinary collaboration, leveraging new technologies, and adopting tailored approaches to tackle the intricate challenges of managing waste. These insights offer valuable guidance for policymakers, waste management professionals, and researchers involved in crafting sustainable waste strategies.

Keywords: solid waste management; optimization techniques; exact models; approximate models; hybrid models; IoT



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1. Introduction

Growing populations, urban development, and economic growth lead to heightened production of solid waste in cities, raising significant issues regarding environmental harm, public health risks, and the depletion of natural resources [1]. The World Bank reports that about one-third of the 2.01 billion tons of annual municipal solid waste is not processed sustainably and predicts this will rise to 3.40 billion tons by 2050, outpacing population growth [2]. Edalatpour et al. [3] stated that the efficient and timely management of solid waste in urban areas is crucial because of the inherent toxins it contains, which pose significant risks to human health, ecosystems, soil, water, and the broader environment. Moreover, solid waste management (SWM) systems' treatment, processing, and transportation-related operations negatively impact the environment, the atmosphere, and society [3]. Addressing

these issues requires the creation and administration of solid waste management systems that are environmentally conscious and consider the interconnections between various activities such as collection, transportation, processing, recycling, and landfilling.

There has been an urgent need for sustainable development over the past few years due to all the issues brought on by the population growth, vast waste generation, insufficient waste management, global warming, and increasingly serious global environmental issues [4].

Waste may take many different forms and can be characterized in a variety of ways [5,6]. Solid waste, originating from industrial, residential, and commercial sources, refers to any discarded material resulting from human or animal activities considered un-wanted and useless. This type of waste can be handled through various management strategies [7]. Landfills are hence often categorized as sanitary, municipal, building and demolition, or industrial waste sites. Waste can also be categorized according to the substance it is made of, including plastic, paper, metal, glass, and organic waste [8]. Moreover, waste that is radioactive, combustible, infectious, poisonous, or non-toxic may be divided into categories based on their potential for danger [9]. No matter where it comes from, what it contains, or how hazardous it could be, solid waste needs to be managed in a systematic way to follow environmental best practices. Thus, environmental planning must take solid waste management into account since it is an important component of environmental hygiene [10]. Then again, the increasing volume of waste and the limited availability of land for waste disposal have made it necessary to adopt innovative strategies to improve the waste management systems' efficiency [11]. Due to the numerous inter-related processes and the highly changeable demographic and socioeconomic aspects impacting the overall systems, waste management procedures include complicated operations and non-linear characteristics [12]. Moreover, it might be challenging to implement SWM systems that work satisfactorily while maintaining other environmental and health standards. Thus, optimization techniques have emerged as a promising solution for optimizing solid waste management systems [13]. Optimization techniques use mathematical models, simulation, and heuristic algorithms to find optimal solutions. To apply these techniques to SWM challenges, a comprehensive review of existing work and findings is essential to encourage further improvements [14]. To date, it appears that there has not been a thorough critical analysis evaluating the application of optimization techniques across different SWM processes.

Thus, the objective of this paper is to provide a comprehensive overview of optimization techniques used in solid waste management and their applications in different stages of waste management processes. The aim of this article is to assist researchers in SWM who are exploring various optimization techniques by discussing key research topics such as the methods employed, their advantages and disadvantages, and their effectiveness. This paper is structured in the following manner: It begins with the Methodology Section outlining the systematic framework used, including the scope of work, research questions, search and selection criteria, paper quality assurance, and data mining strategy. Subsequent sections delve into the major optimization techniques identified in the survey, followed by a detailed discussion of the various fields within solid waste management where these techniques have been applied. The paper concludes by addressing the limitations and challenges encountered in implementing optimization techniques in solid waste management and offers recommendations for future research.

2. Methodology

This section outlines the methodology followed to identify relevant studies and filter them before examining the models used. The key goal of the systematic literature review (SLR) was to establish an unbiased review method, which led to comprehension of outcomes and credibility. SLRs were used to find, assess, and analyze studies in a certain field of study [15].

2.1. Systematic Review Protocol

The methodology for this paper followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020, see Supplementary Materials) to specify the methods used to perform the analysis by categorizing the process into four phases [16,17]. The flowchart outlining the suggested technique is shown in Figure 1, where PRISMA 2020 is applied in four stages. The first stage involved identifying review characteristics, which establishes the scope of the research. Thus, the SLR's primary purpose was to detect and assess published research that addresses the application of optimization techniques in SWM. This could be accomplished by addressing three main questions:

1. What studies have been conducted on optimization techniques in SWM?
2. How is the performance of various optimization models in SWM?
3. What are the advantages and drawbacks of applications of optimization techniques in SWM?

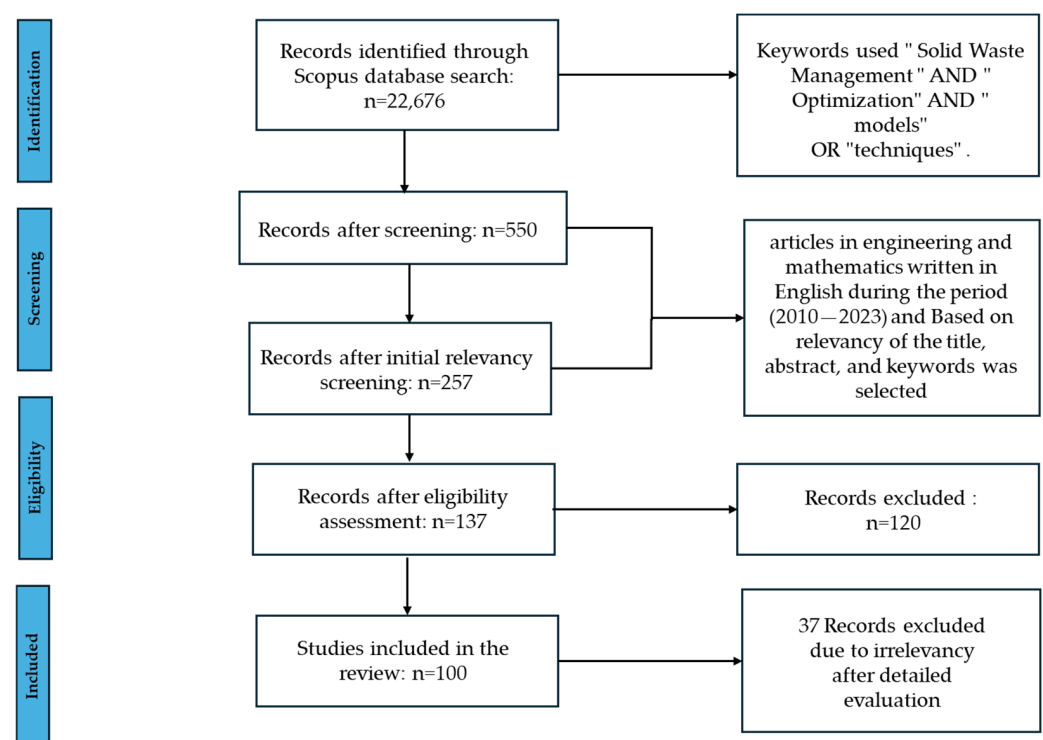


Figure 1. Systematic review methodology by PRISMA.

The screening process constituted the second stage of the PRISMA approach, involving the selection of databases and the careful choice of search terms and keywords to identify articles relevant to the topic. The analysis of eligibility was the third stage, where we started by conducting an abstract review, in which abstracts were reviewed and assessed to decide whether the publication fit within the scope of the literature review. Based on that, papers that did not fit the scope of the review were excluded. If the abstract aligned with the research scope, the full paper was then reviewed to highlight key points and contributions and to collect additional relevant papers and references using a snowball, backward referencing method from this study's bibliography. Finally, the fourth stage of PRISMA was the synthesis and assessment of collected data. This was accomplished by first classifying the articles gathered in accordance with the publication date and location, which were regarded as the first stage in the data analysis procedure. Following that, the chosen papers were divided into four groups relative to the optimization procedure: exact models, approximate models, hybrid models, and IoT-based models. This approach was beneficial as it provided a general overview of optimization models employed in SWM, pinpointed critical features that future research should focus on when developing

Table 1. Breakdown of studies by country.

Country	Number of Articles	Citations
China	17	284
Australia	6	279
Iran	8	210
Spain	11	207
United Kingdom	10	234
France	6	110
USA	14	252
Brazil	5	99
South Korea	5	58
India	7	66
Italy	8	166

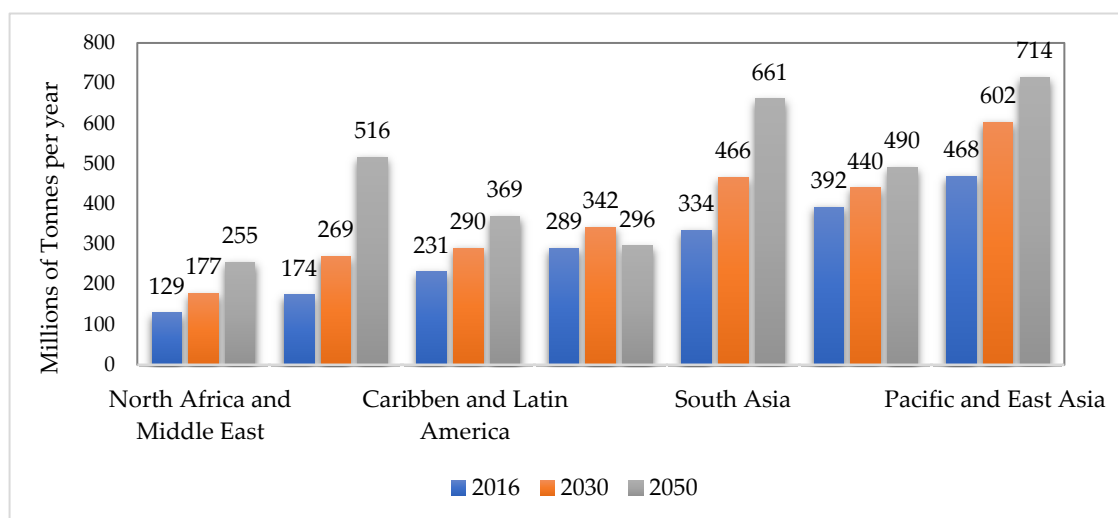
3. Results and Discussion

Based on the systematic review conducted, the definitions of SWM, optimization models, and solution procedures considered in the literature are discussed in this section.

3.1. Solid Waste Management

Solid waste management is an integral component of urban planning, comprising a variety of techniques for dealing with waste generated by human activities. Effective solid waste management procedures strive to protect public health, decrease pollution, and promote resource sustainability [20].

Several countries, including the US, China, Canada, Malaysia, and the Philippines, as well as Middle Eastern and many European countries, have experienced increasing waste generation, as shown in Figure 3 [2]. To better comprehend the issue and provide appropriate disposal options or halt this rise, various governments have started conducting simulations to predict waste amounts [21]. In different nations, there are differences in how waste is managed and where it ends up. There are nations that are more aware of the value of recycling and employ technical and legislative means to boost recycling. Germany, Holland, Denmark, and Belgium, for instance, are highlighted in the recycling of building waste [22].

**Figure 3.** Solid waste generation through years [2].

Solid waste management encompasses several stages, beginning with waste generation, followed by collection, storage, transportation, treatment, and, ultimately, disposal, as depicted in Figure 4 [23].

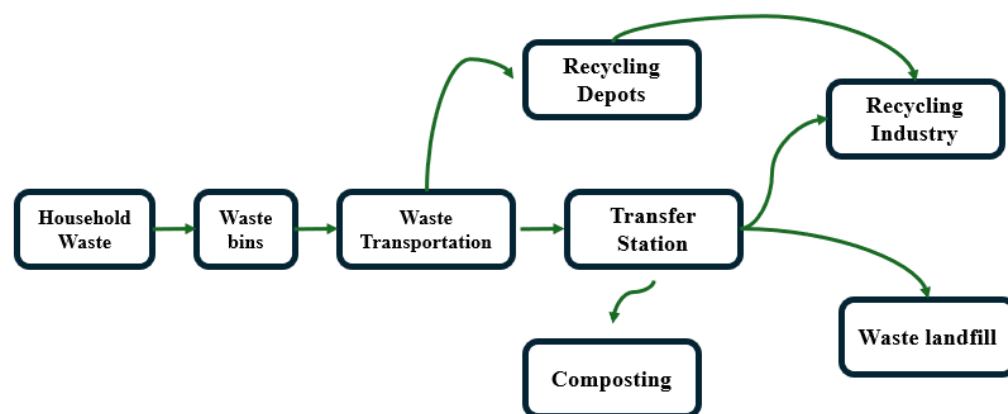


Figure 4. Solid waste management cycle [9].

For waste generation, optimization is essential to prevent, or minimize, the quantities of generated waste and hazards correlated to waste generation [24]. Moreover, Damamy [25] indicated that in the waste collection stage, methodologies of collection differ depending on the facilities, and thus, improving the process depends on many factors such as collection method, levels within facilities, and so on. For waste storage, incompatible waste should not be mixed or brought into touch with one another during waste storage. This also enables the examination of containers to check for leaks or spills. Examples include allowing enough space among incompatibles or using barriers like walls or containment curbs to physically separate different waste types [23]. In addition, in the transportation stage, waste should be transported both on-site and off-site to enable processing and avoid or minimize spills, discharges, and exposures to workers and the public. Any waste containers intended for shipping away from the site must be secured and labeled with their contents and any potential risks [26]. The treatment of waste comes next, which can be achieved by recycling and reusing waste. Recycling programs may be put in place, in addition to waste avoidance techniques, to greatly reduce the overall quantity of waste [27].

Optimization techniques can be used to help ensure compliance with the SWM regulations that have been put forth by governments. Utilizing optimization algorithms, a decision support system can be created. Several sorts of stakeholders, including public managers, business owners, regulatory agencies, and environmental managers, are involved in solid waste management. Certain environmental issues must be resolved quickly, as several stakeholders may need the processing of a sizable quantity of data, and the offered solutions must be open and logical. Many procedures involved in solid waste management might lead to issues with planning, control, logistics, and recycling. It involves a multidisciplinary approach that integrates the three pillars of sustainability and incorporates multi-criteria evaluations throughout every phase of its lifecycle. Research is being conducted in several areas, including the forecasting of waste creation, container transportation management, container collection system monitoring, and the construction of new waste disposal facilities [28].

3.2. Optimization Techniques in SWM Based on the Model Type

Analysis of previous studies showed that there are many models used to optimize different parameters of solid waste management through all phases of the process. This section provides a comprehensive review of different models within the four main categories obtained from the analysis. Most of the previous studies indicated that mathematical programs are useful tools for deciding on the optimal cost strategies for MSW management problems. The following subsections discuss different models used for optimization purposes in different stages of the SWM process.

3.2.1. Exact Models

Exact models or deterministic models are the most used mathematical models in SWM [29]. Such models can determine an exact optimum solution for a mathematical program. Linear programming (LP), pure and mixed integer programming, multi-objective programming, and dynamic programming are some of the methods used for deterministic models. More details on each type of these mathematical formulations are presented below.

A. Linear Programming in SWM

In convex optimization, the point of local optimality is the same as the point of global optimality, encompassing linear optimization models as a particular instance [30]. Contrarily, general non-linear optimization concentrates on techniques for identifying local optimum solutions. When solving generic non-linear models, there is always the possibility that the solution reached may not be the global optimum. It follows that a tendency towards LP models in SWM investigations is unavoidable [31]. Thus, several studies that considered solid waste management proposed an LP model that deals with different processes or stages in SWM [32]. In an LP formulation, the goal is to either minimize or maximize a linear objective function, constrained by linear inequalities and equalities [33]. Many of the current linear models used in SWM are designed as mixed integer-linear programming (MILP) models with a single objective function. These models often include assumptions that restrict their applicability to municipalities and countries [34–36]. Other researchers [37,38] have developed multi-objective mathematical models. One study [39] suggested a MIL model for the optimization of waste vehicles routing problems, where the weighted sum of the fleet size was modeled as the objective function to find the minimum operational cost. Another study [40] argued that the efficiency of waste collection depends on pins location; thus, a MILP model for bin allocation was developed. Moreover, Yang et al. [41] proposed a dual linear fractional (LF) model to support urban–rural SWM in Xiaman, China. The developed model can maximize the system efficiency while minimizing the cost. Using this model, the issue of congested landfill was resolved by determining the ideal waste flow for each facility. Also, solutions were offered for waste allocation and the expansion of facility capacity. Similarly, a MILP was developed to find the optimal location for Polyethylene Terephthalate PET waste processing in Mexico, as it is considered to be the third largest consumer of PET [42]. A multi-source, three-echelon location problem was recognized involving the transfer of generated PET waste in five towns in Veracruz region, Mexico. This waste is directed through a selected collection center to satisfy the requirements of three demand points in the reuse market. This approach was implemented with the aim of minimizing the network's overall cost, which was established as the objective function.

Moreover, Munguía-López et al. [43] proposed a mathematical model aimed at optimizing SWM by developing a circular value chain. This model was specifically tailored to process various kinds of COVID-19 medical waste within a modernized industrial park setting. The problem was formulated as a MILP to maximize revenues while reducing the environmental effect by reducing untreated waste amount. New York City was the case study to apply the proposed technique, where there has been a large rise in the output of medical waste. The mathematical model consisted of 341 equations, 63 binary variables, and 317 continuous variables. Varying tax rates were suggested based on the volume of waste deposited in landfills to encourage recycling. The results were displayed on a Pareto curve, illustrating the balance between profits and waste processing. The analysis showed that taxation motivates recycling, including of materials that are not economically advantageous to recycle, thus enhancing profits and reducing the environmental and health risks related to medical waste.

Additionally, Shaban et al. [44] developed a MILP model for SWMS configuration that combines waste producing sources, stations for collection/transfer, recycling facilities, burners, and landfills. The suggested model was designed to estimate the number of the various facilities and their optimal locations, as well as the best waste flow in the system, to reduce the system's daily cost. The objective function is subjected to 30 constraints to define

different decision variables and ensure the amount of generated waste equals the waste collected and transferred along with other constraints. Fayoum Governorate, Egypt, was selected as the case study to test and analyze the model. The primary contribution of this study was the theoretical development of a MILP model intended for the optimal design of SWMSs in developing nations, along with its operational implementation as demonstrated in the SWMS of Fayoum Governorate, Egypt. A similar system was developed by Xie et al. [45], where the research investigated the optimization of the SWM supply chain (SWMSC), where it consists essentially of waste separation and waste transportation from collection locations to transfer and treatment centers.

Šomplák et al. [46] established a MILP model to reduce the waste production amount, ensure the highest recycled amount, and benefit from the residual waste for energy recovery. Their approach incorporated both pricing and marketing principles into the MILP model, considering the evaluation of greenhouse gas (GHG) emissions and the minimization of costs. Their goal was to create the optimum WM grid to symbolize a sustainable economy with eco-friendly considerations. The suggested multi-objective model was used to facilitate decision-making at the micro-regional stage in the Czech Republic as a case study in the domain of waste treatment infrastructure design. The use of circular economy concepts, considering the whole quantity of GHG produced, highlighted the existing promise in waste prevention. Alternatively, there is a limited increase in recycling, landfills are not promoted, and the recovery of energy is desired. However, the decision-maker is responsible for the complex system's planning. Another MILP model to find optimal the recycling process for MSW was developed by Sadrnia et al. [47], where the objective function is to find the minimum recycling process cost. On the other hand, Hao et al. [48] formed a multi-objective model for site allocation to optimize the waste collection cost, environmental impact, and GHG emissions. It was shown that including environmental impacts and GHG emissions enabled an environmentally friendly model output. Książek et al. [49] made a significant contribution by formulating an optimization challenge aimed at equitably distributing travel distances among the heterogeneous waste trucks within a fleet. The problem is designed to meet the demands of Municipal Solid Waste Management (MSWM) in Cracow, Poland, where the objective is to achieve a harmonized working time for routes while minimizing the overall duration of the collection service. To address this, an MIP was formulated to find optimum team schedules. The goal was to efficiently serve the existing network of solid waste pickup points using a heterogeneous fleet, with the added constraint that up to 30% of the fleet consisted of electric trucks. On the other hand, Pouriani et al. [50] introduced an innovative MSWM network designed to minimize diverse costs. A bi-level MILP model was obtained, with the lower level addressing the costs of establishing collection centers and their location and the upper level focusing on waste allocation across various centers. To address the inherent uncertainty in the quantity of waste collected, a scenario-based optimization method was incorporated. The efficacy of the suggested model was demonstrated through a case study conducted in Babol, Iran. The findings reveal that the strategically selected collection centers are situated in regions with minimal distance from their respective coverage areas, optimizing the flow of waste/products.

Furthermore, Asefi et al. [51] focused on the exploration of an integrated framework addressing the Mix Vehicle Routing Problem (VRP) and fleet size with the overarching goal of optimizing a cost-effective integrated SWMS. A pioneering bi-objective MILP model was developed, aiming to simultaneously minimize transportation costs across the entire network and the total difference from equitable capacity allocation to transfer centers. The scope of their investigation encompassed an integrated solid waste management (ISWM) system, including interdependent facilities and incorporating diverse technologies. The problem was characterized by a heterogeneous fleet subject to waste compatibility constraints and various technologies. This holistic approach contributes to the advancement of optimal solutions for complex ISWM systems by considering various interdependencies and constraints within the waste management landscape. The same authors [52] proposed

a MILP for optimizing a logistic network and transportation system for the ISWM system. To tackle uncertainties in MSW ratios of generation, the study proposed a two-phase stochastic optimization method. This approach efficiently supports a cost-effective ISWM system for transportation by evaluating optimal fleet size, capacity allocation, and vehicle routes. The same issue was considered by Wu et al. [53] in Tiwan, who proposed a MILP model to enhance service provided for all residents. However, many constraints need to be considered to improve accuracy of the problem formulation.

Moreover, Mohammadi et al. [54] proposed a MILP model for the efficient utilization of MSW within a supply chain network. The approach focused on supply, production, and distribution choices at the strategic and operational levels. Utilizing the best waste-to-energy tools while taking capacity, environmental, and market demands into account, it pursued the optimization of the yearly net profit. The suggested sustainable management approach not only reduced the environmental impact but also efficiently delivered solid waste throughout the system and turned it into power, therefore assisting in the production of energy. This study was different from others as it focused on waste-to-energy technologies to find the optimal sustainable MSW handling and processing method.

In summary, the fundamental distinction among the different linear and MILP models for MSW optimization is the precise objectives that they aim to achieve. Some models attempt to reduce costs, while others aim to reduce environmental consequences, increase recycling rates, or fulfill specific waste diversion objectives. Also, the constraints can vary significantly depending on the specific problem formulation, including waste collection and transportation capacity, recycling facility capacities, landfill space limitations, and environmental regulations. Linear and MILP models are very useful in MSW optimization due to their ability to address various objective functions and constraints. However, their complexity can be difficult, and the choice of specific model formulation must align with the waste management organization's goals and resources.

B. Integer Programming in SWM

The pure integer programming application in SWM is very limited as per the conducted search and criteria followed in this paper. Integer programming (IP) is applied when the decision variables are restricted to integers only [55]. A previous study [56] presented a model to assist the MSW system's decision-making process. The goal is to move from a system based on a door-to-door approach to a waste bins system, which is anticipated to lead to effective logistic expenses. This model tackles the issue of choosing the ideal location of the waste bins for a specific city in Argentina. The objective is to find garbage accumulation points (GAPs) while optimizing two distinct factors related to system costs. The first requirement is to reduce overall investment costs, or the price of each individual bin. Enhancing the GAPs' "autonomy" is the second requirement. The autonomy of a GAP is determined by how many days can pass between two collections (to empty the bins) by the collection truck. This model was applied to actual scenarios that included the collection of unsorted waste, which was the existing state in the city and could, thus, be easily implemented. In addition, some scenarios included waste classified as dry and humid, which could be introduced once the community was accustomed to using the waste bins. The suggested attempt identified a set of possible results for all scenarios. Also, Braier et al. [57] considered an IP to enhance the recycling garbage collection system in a sizable town in Argentina. The produced solutions for the optimum route outperformed the previous routes that were designed manually, with 100% of the municipality's blocks covered by the model solutions, as opposed to up to 16% with the manual routes.

A similar problem of selecting the sites for waste collections was introduced [58]. The study proposed an IP model to help determine the placement of waste bins in a town and the number of bins required at each location. The model aided in evaluating tactical choices by imposing constraints, where each collection area is large enough to accommodate the garbage that will be sent there, while also taking into consideration quality of service restrictions from the perspective of the residents. Moreover, Gallo et al. [59] developed an IP model to enhance the waste collection management system by finding the optimum location

of waste transfer stations. Another IP model was generated by Rambandara et al. [60] to find the optimal route for the waste collection process. A similar study was conducted by Zhang et al. [61], who proposed a robust model to address the complexities of the multi-trip transportation and collection of MSW in an uncertain environment. Residents' satisfaction is quantified as a penalty charge relative to time window constraints. A case study was conducted using real-world data from MWS transportation and collection in the District of Beijing, China. The CPLEX program was used to validate the solution. Additionally, a sensitivity analysis on the related parameters was conducted, exploring the effect of work hour limits and time windows on the service levels and the total costs.

In general, the application of linear and integer programming techniques is a typical trait of optimization models for SWM. Most models rely on a streamlined mass flow modeling method that focuses solely on the waste stream fluxes, like residual waste, rather than the waste materials themselves. Therefore, models are unable to consider temporal and regional variations, as well as changes in the streams' compositions following treatment, which have a significant effect on the economic and environmental performance of the SWM system. Therefore, non-linear programming models are necessary to account for the waste streams' diverse, variable, and varied nature. The following section discusses the non-linear programming optimization models developed for SWM.

C. Non-linear Programming in SWM

Non-linear programming (NLP) techniques can be utilized to optimize waste disposal costs and resource utilization in SWM problems, as SWM systems involve multiple non-linear objectives, constraints, and decision variables [62]. Some examples of non-linear relationships in SWM include the relationship between waste generation rates and economic growth, the relationship between waste disposal costs and distance to landfills, and the relationship between recycling rates and the availability of recycling facilities.

For example, Araya-Córdova et al. [63] identified the optimum combination of two choices for allocating current resources to meet the challenge of optimal resource allocation for recycling system adoption by municipalities in rural and urban regions of a developing country. After comprehensive data collection, a non-linear optimization model was developed to propose a recycling policy based on the combination of two choices, increasing and reallocating existing MSWM resources, which maximizes the average value of the probability of municipalities adopting a recycling program.

Another non-linear model was introduced by Roberts et al. [64]: the model utilizes an evaluation of both environmental and economic impacts associated with the treatment of produced waste at existing facilities. These impacts are determined using a series of lifecycle process models, which employ non-linear equations tailored to each type of waste material and treatment method. It was concluded that the willingness of a waste producer to engage in recycling, composting, and other waste treatment procedures has a substantial impact on various waste collection and treatment processes. The degree of waste flow contamination is affected by the producer's behavior; this is likely to differ regionally, chronologically, and demographically. Due to the heterogeneity of the waste flows and their dynamic and different compositions, non-linear models are necessary for each procedure and material. This creates non-linear restrictions, considerably increasing the model's complexity.

Moreover, Zhao et al. [65] presented a mathematical programming model to address MSWM. The model was designed as a mixed integer non-linear programming (MINLP) model for scenarios involving variable capacities at treatment facilities and as a mixed integer linear programming (MILP) model for situations with fixed treatment plant capacities. The study considered different scenarios: the lowest cost with/without adjustable capacities, the lowest carbon emissions with/without adjustable capacities, and the highest carbon emissions with/without adjustable capacities. The findings summarized the cost and carbon emission for the different scenarios. These findings aided decision-makers in developing effective MSW management policies. The study presents an efficient strategy for managing MSW in other developing nations.

Another utilization of non-linear representation was studied by Nevrlý et al. [62]. The study examined the link between GHG emissions and the treatment cost of MW, as well as the environmental effect of different handling strategies. To maximize future MW treatment solutions across a vast geographic region, an MINLP model was formulated. The use of a non-linear model was justified based on the non-linearity of waste treatment costs and incentives related to reduced GHG emissions. The Czech Republic was selected as the case study, and the obtained outcomes suggest a promising reduction in the emissions of GHGs by approximately 150% and an increase in waste treatment costs of only approximately 2.5 EUR/ton.

Additionally, Rizwan et al. [66] created an MINLP model to find the optimum MSW processing routes while considering two different MSW management scenarios. The MINLP was converted into its corresponding MILP form for simplicity of its solution. The optimization problem solution offered the best method for creating usable products from municipal solid waste, guided by the specified economic objective function. The created methodology was employed in a case study in Abu Dhabi to determine the best route for MSW's conversion into value-added products and energy. The findings suggested that a combined path that produces bioethanol from the remaining waste through gasification and catalytic transformation, while recycling the MSW, has the potential to be economically advantageous. Additionally, a sensitivity analysis was conducted to determine how important technical and economic parameters affect the optimization outcomes.

Also, Yadav et al. [67] discussed the problem of determining economically optimal locations of an MSWM infrastructure unit such as waste transfer stations in India. The heterogeneity data collection methods, measuring the distance on the road and strategically allocating transfer stations, were considered when proposing the optimization model. Thus, the study developed an MINLP as the model for searching for the best location option for the waste transfer station. The constraints are the source of non-linearity in this model due to the heterogeneity of the data collected. Furthermore, Rathore et al. [68] investigated the viability of a circular economy approach in MSWM, considering economic, environmental, and social aspects. The proposed concept involves converting collected organic MSW into biogas, which is then utilized as a fuel in a thermal power plant, thus reducing dependence on coal mining. An MINLP model was generated to minimize the total cost, comprising functioning, transportation, hiring, environmental, social, and penalty costs. The model was applied to different scenarios, revealing that the suggested system outperforms the existing ones by effectively reducing the overall cost.

The strengths of these studies include advancements in optimization strategies and dynamic system understanding. However, potential drawbacks may involve challenges in practical implementation and the requirement for extensive data inputs, especially in complex modeling scenarios.

D. Dynamic Models in SWM

Dynamic modeling is usually utilized to explain and forecast how different parts of any system will interact over time [69]. In SWM, it has been shown that in comparison to a static policy with defined collecting and hauling lengths, there are dynamic routing and scheduling strategies that have minimal operational costs. Thus, many waste collection firms use routes and scheduled pick-up times and dynamic optimization of waste collection [70]. Another application of dynamic models was utilized [71], where a dynamic MIP model was generated to find the optimal planning of SWMSs, with the waste being collected from different cities and transported to different processing sites. The results showed that waste disposal and collection costs depend on time. Accordingly, the cost resulting from the dynamic model was less than the cost calculated when using non-dynamic models.

3.2.2. Approximate Models

According to Cheng et al. [72], fuzzy mathematical programming (FMP), interval linear programming (ILP), stochastic mathematical programming (SMP), and grey system theory

are methods used in approximate models of SWM. The approximate models included in this systematic review are discussed in the following subsections.

A. Stochastic programming models in SWM

Due to the complexity of and interactions in SWMSs, certain system characteristics should be treated as random variables for greater accuracy. Thus, these systems must be modeled stochastically using data analysis and human judgment. Consequently, many approximate optimization models have been created to address SWM issues under uncertainty [73]. One model is the stochastic chance-constrained programming (SCCP) model [74], which was developed to deal with bi-random variables. It was applied in SWM, where a model for optimizing the SWM systems under uncertainties was developed [75]. Similarly, a stochastic MILP (SMILP) model with two stages was produced to determine the optimum annual cost, material distribution, waste treatment technology, and capacity of treatment facilities in a MWS supply chain system [76]. The model used an L-shaped decomposition algorithm to find the solution as it is commonly used in two-stage stochastic problems. Another application of the SCCP model was introduced by Zaeimi et al. [77], who developed a model where the constraints considered parameters' uncertainty. The suggested model is a MILP for the minimization of the total cost and pollution for a waste collection process. The findings showed that minimizing uncertainties led to significant reductions in cost. Moreover, Gambella et al. [78] produced a stochastic model for optimal SWMS processes. The effects on the solution due to stochastic waste production have been studied, and the model demonstrated the advantages of the stochastic approach over the deterministic approach, which can result in an inaccurate decision plan. Also, Xiong et al. [79] suggested a multi-phase stochastic LP model to optimize the probability of each waste-to-energy facility meeting its financial objective while considering the environmental constraints in a city in Singapore. The model obtained the optimal balance of different treatment procedures that is more practical than existing ones.

While these studies share the common goal of optimizing waste management under uncertainty, differences arise in their geographical foci, specific waste types, and modeling approaches. Strengths lie in their contributions to advanced optimization methodologies tailored to diverse waste management scenarios. The drawbacks may include potential challenges in practical implementation, data availability, and the complexity of stochastic modeling, highlighting the need for the careful consideration of regional variations in waste management practices.

B. Fuzzy programming Models in SWM

Another representation of uncertainties in SWM is fuzzy programming models, especially in environment management systems. One application was presented by Wang et al. [80], who developed a stochastic fuzzy programming model to manage a long-term SWM system. This study concluded that the proposed model improved the solution as it dealt with highly uncertain constraints. Also, Huang et al. [81] presented a stochastic-fuzzy quadratic programming (SFQP) model to find the optimum MSWM policy. This model was selected because of the complexities within the system under study, including system components dynamic variation, uncertainties in SWM facilities, rates of waste generation fuzziness, policy examination for allocating waste streams, economic and environmental objectives, and demands for waste diversion. The solutions devised helped to generate multiple environmental and economic policies. Also, Govindan et al. [82] produced a bi-objective MILP model for handling medical waste generated during the COVID-19 pandemic. The suggested approach effectively reduces both the overall expenses and the hazards associated with the public's contact with pollution. The model was solved using a fuzzy goal programming approach; then, the effectiveness of the model and its solution strategy were assessed using data from 13 medical waste generation points in a region west of Tehran, Iran. In a related context, a linear programming model was considered to discover the most secure and fastest routes for the trucks during the pandemic [83]. The model was tested by Istanbul's health facilities and was applied to a specific area, where

recommendations for routes for medical waste vehicles have been based on the results. This model has a very adaptable structure, allowing its adoption in a variety of locations and sectors.

Moreover, Srivastava et al. [84] focused on choosing the optimal treatment and disposal centers, capacity planning, and allocating trash while considering uncertainty in long-term SWM planning. The proposed non-linear model was designed as a multi-period, multi-objective system for integrated SWM planning. The model dynamically assigns waste to facilities while considering the capacity of handling facilities and the quantity of waste. Additionally, the model addresses uncertainties in both the volume of waste generated and the operational capabilities of treatment facilities by incorporating fuzzification. It was discovered in the study that variations in waste quantities impact the planning of waste treatment and disposal facilities more significantly than uncertainties in their capacities. It was concluded that the link between rising waste volume and rising waste management costs and risks is non-linear. Therefore, even slight fluctuations in waste quantities can significantly raise the overall costs or associated risks. The results of the research can be used to comprehend how changes in the priorities and goals of planning decisions affect the choice of facilities and waste diversion. Similarly, Guo et al. [85] incorporated fuzzification into the constraints, creating an inexact stochastic-fuzzy quadratic programming (IFSQP) model to efficiently allocate waste to available facilities within a non-linear framework. Their objective was to achieve optimal waste flow throughout the course of the whole planning scope to reduce the overall estimated system cost. The constraints included all connections among decision variables, waste generation rates, waste diversion goals, and the capacities of waste management facilities. The findings showed that in many scenarios, reasonable solutions were achieved, and the proposed method stands out when compared to other techniques that deal with uncertainties due to its unique special characteristics. However, Srivastava et al. [84] emphasized the importance of considering a broader spectrum of uncertainties.

Furthermore, Chang et al. [86] presented a fuzzy multi-objective LP model to identify an optimal compromise strategy for MSWM. The objective was to find the minimum operating costs and air pollutant emissions. In contrast to previous research that primarily focused on either economic optimization or environmental impacts, this approach considered both aspects. The findings show that the best balance of a MSWM strategy not only generates a net profit but also aids in decreasing emissions of air pollutants. Moreover, scenario analyses of recyclable rates highlight the dual benefits of resource recycling, positively impacting both economic optimization and the minimization of air pollutant emissions.

The studies and applications outlined offer a comprehensive overview of the diverse and sophisticated use of fuzzy programming models in SWM. Fuzzy programming models significantly enhance SWM by adeptly navigating uncertainties in environmental systems. They optimize waste management strategies, balancing economic and environmental goals, and facilitate adaptable solutions across various scenarios. These models prove crucial for efficient, sustainable waste management and policy development amid complex, uncertain conditions.

3.2.3. Hybrid Models in SWM

A substantial increase in interest in hybrid models has been witnessed in recent years. These models combine the advantages of several distinct approaches to produce effective solutions to optimization problems [87]. The combination of genetic algorithms with deterministic models represents one of the well-known hybrid genetic algorithm models that is currently used by researchers to optimize SWM. Genetic algorithms (GAs) are defined as optimization approaches that mimic the process of natural development [88]. They are stochastic algorithms designed to simulate specific natural processes and are considered one of the exhaustive search approaches in artificial intelligence [89]. One use of hybrid models was proposed by Moazzeni et al. [90], where a dynamic model for electric waste collection vehicles was developed to find the optimal collection vehicle route plan and

choose the best sites for waste collection facilities and charging stations. The study also used GA to generate a solution to the problem. Another study [87] generated a stochastic chance-constrained programming (SCCP) model to investigate two sub-models. The first sub-model routes the fleet among waste-generating and -separating facilities using the VRP, and the second sub-model considered resources' distribution from separation centers to collection or landfill centers. The goal is to enhance operational efficiency by accounting for the unpredictable factors in the output of separation facilities and the recovery value of each bin. The proposed model aims to maximize recycled income while reducing overall transportation costs. Another interesting study developed a MILP model to optimize the operational cost and CO₂ emissions for the waste collection process from generation source to separation centers. The optimal solution was generated using metaheuristic algorithms and various new heuristics [91]. The performances of the suggested algorithms were assessed to rank them according to the relative percentage deviation and the relative deviation index. The analysis discussed the Social Engineering Optimizer (SEO), which is a newly established metaheuristic algorithm, and the simulated annealing (SA) algorithm, which is usually used with VRP to resolve complex optimization problems based on the similarity between the physical behavior of metal annealing and an optimization problem [92]. Another study [93] developed a MILP model to optimize the waste collection process. To find the optimum cost-effective and environmentally friendly solution, the research used one metaheuristic algorithm that consists of an adaptive variable neighborhood search method (AVNS) and construction heuristic. The results showed that AVNS is a highly effective algorithm for electric waste collection problems. Moreover, Tirkolaee et al. [94] presented a novel MILP model to optimize the fixed/variable costs and GHG emissions for the MSW collection, transportation, and disposal system. To effectively identify the solution, the study utilized a hybrid metaheuristic algorithm that integrated a multi-objective invasive weed optimization algorithm (MOIWOA) and a multi-objective simulated annealing algorithm (MOSA). In the mechanism of MOIWOA, the initial population is a set of weeds, which first locate ideal farmlands to settle on, and only then do they start new colonies. Utilizing this behavior allows the process to tackle optimization problems. The hybrid MOSA-MOIWOA algorithm outperformed classical metaheuristic algorithms.

Moreover, the farthest-candidate heuristic (FCH) method was applied to determine the optimal solution for the non-linear MIP model proposed for the multi-objective SW collection problem [86]. The solution process used CPLEX to evaluate the heuristic results and the comparison confirmed the efficiency of the heuristic method. The FCH is based on the sampling theory and best-candidate sampling technique, where all boundary points are first identified and then the candidate solution closest to the newly chosen point is chosen for each new candidate solution [95]. Also, Rossit et al. [96] generated a MILP model to optimize waste bin locations. The solution algorithm utilized PageRank algorithm-based heuristics to find the near-optimal solution, where the problem is defined over a weighted network using the PageRank algorithm. In such networks, the peaks represent potential sites, and the connections indicate the importance or weight of each linked vertex. The vertices are then represented by a vector using a function that sorts them according to the importance of each vertex in the entire system. The entire system is then configured by iterating through the vector of sorted vertex data using a constructive heuristic method. It showed an exceptional performance compared to other heuristics. Mirdar et al. [97] presented a sophisticated multi-phase MILP model crafted to enhance the efficiency of a Sustainable and Integrated Recycling and Disposal Network (SIRDN) for MSW. The model uniquely integrates environmental, economic, and social considerations, thereby comprehensively addressing the spectrum of sustainability. The primary objective of optimization lies in maximizing profit. Another study [98] investigated an innovative heuristic procedure, incorporating both an IP model and ant colony optimization (ACO) to formulate an optimal two-shift collection plan that incorporates Resilience, Accessibility, and Connectivity (RAC) factors. The model was designed to identify strategically located collection points during each shift, emphasizing proximity considerations. Subsequently,

the ACO algorithm was employed to delineate the most efficient plan of routing for each shift, considering the dynamic interplay of RAC factors. Through the consideration of a case study focused on a Taiwanese city, the results established the proposed collection plans in comparison to existing ones, as evaluated based on both proximity metrics and collection distance.

From the discussed research on hybrid models, it can be concluded that hybrid models have revolutionized SWM by blending genetic algorithms with deterministic and other optimization techniques, leading to more efficient, adaptable solutions for complex SWM challenges. These models excel in routing optimization, cost reduction, and enhancing recycling processes, demonstrating superior performance over traditional methods. By employing metaheuristic algorithms and innovative heuristics, such as the Social Engineering Optimizer and adaptive variable neighborhood search, hybrid models offer practical, environmentally sustainable solutions, underscoring their significance in advancing SWM towards sustainability.

3.2.4. IoT in Solid Waste Management

According to Marques et al. [99], a stunning modern technology identified as the internet of things (IoT) offers potent ways to update outdated systems. IoT technologies can play a significant role in SWM by improving the efficiency of waste collection, reducing waste overflow, and promoting sustainable waste disposal practices. IoT-enabled smart bins detect when they are full and alert the waste management team to collect the garbage. These smart bins can also provide real-time data on the level of waste inside them, which can help the waste management team plan collection schedules more efficiently. One study [100] developed a model for routing and allocation using IoT-based smart bins. The model used a combination of an ant colony optimization method (VNS-ACO) and intelligent variable neighborhood search, along with a hybrid metaheuristic algorithm, to find the problem solution. The performance of the proposed hybrid approach surpassed classical algorithms. Another IoT application in SWM was introduced by Pal et al. [101]. The application utilized the citywide placement of the IoT-enabled bins to provide input data and monitor the volume of garbage in each bin. To minimize SWM's total cost, the suggested system offers an efficient route for the collection trucks and recommends the capacity of the collection vehicles according to the geographical location. Additionally, the proposed approach enhances smart SWM's performance by extending the lifespans of IoT devices. Also, Mishra et al. [102] utilized IoT-based bins to optimize the resources needed for the waste collection process. The novel cost-function-based route optimization technique was concluded to be quite efficient in vehicle route problems.

IoT technology integration in SWM has been transformative, significantly enhancing the sustainability and efficiency of waste collection and disposal practices. IoT-enabled smart bins, capable of signaling their fill level, enable precise and timely waste collection, thereby reducing overflow incidents and optimizing collection routes. Studies employing IoT-based models for smart bin allocation and routing have shown that hybrid metaheuristic algorithms, like the combination of variable neighborhood search with ant colony optimization, outperform traditional algorithms. This innovative approach enhances the waste collection process while also promoting the cost-efficiency and environmental sustainability of SWM.

3.3. Optimization Techniques Based on SWM Processes

This section discusses the optimization techniques based on SWM processes: waste collection and transportation, waste processing and recycling, and those used in integrated SWM supply chain processes. Tables 2–6 summarize some of the optimization models in the different stages.

The main findings from the reviews of various SWM models in the collection stage include strategies to minimize waste collection costs, optimize bin allocation, and select the best locations for collection centers (Table 2). The review spans exact models (MIP, IP, NLP), approximate models (SCCP), and hybrid models combining genetic algorithms (GA) with

other approaches, highlighting innovations like the use of IoT for smart bin allocation and the PageRank algorithm for fast solution generation. Key limitations and future research opportunities identified include the need to consider more complex scenarios, uncertainties in demand, and the effects of different types of collection vehicles and waste. Moreover, the potential for using fuzzy programming, robust optimization, and dynamic modeling to enhance these models is noted, alongside the importance of considering social and environmental impacts, such as GHG emissions.

Table 2. Optimization models in SWM—the collection stage.

Model Category	Model	Main Contribution	Limitations and Future Research Opportunities	Ref.
Exact Models	MILP	A model that effectively minimizes the cost of waste collection	More complex cases need to be considered	[39]
		A model for optimum bin allocation		[34,36,40]
		A model to find an optimum collection time for a heterogeneous fleet		[35]
		A model to select optimum location for collection center	Random demand must be considered	[42]
		To find the optimal crew schedule using a heterogeneous fleet	Add more types of collection vehicles to investigate their effects on the model	[49]
	IP	A model to optimize the waste collection cost	More complex cases need to be considered	[50,56,57,59,61]
		A model to select optimum location for collection center	Consider changing filling rate of the waste bins	[58]
	NLP		Select an appropriate algorithm to solve and consider uncertainties	[67]
Approximate Models	SCCP	A bi-objective model that considered parameters' uncertainties	Fuzzy programming and robust optimization need to be investigated	[77]
Hybrid Models	GA + SCCP	A model to find optimal routes from generation to separation facilities	Environmental and social facets of GHG emission need to be considered in the model	[103]
	GA + Dynamic model	A model to locate the charging stations with and routing vehicles	Results can be duplicated for other countries and different MSW systems	[90]
	Heuristics/Metaheuristic + MILP	A model where different heuristic and metaheuristic algorithms are used	Introduce dynamic modeling to the problem	[91]
	AVNS + MILP	A model generated by Using an electrical fleet to reduce transportation costs and CO ₂ emissions	Results can be duplicated for other countries and different MSW systems	[93]
	ACO + IP	A model that concludes that the ACO is efficient in routing problems	Many assumptions were made to simplify the model, allowing more complex cases to be considered	[98]
	IoT + (VNS-ACO)	A model based on smart bin allocation by IoT and vehicle routing, using an intelligent hybrid VNS-ACO algorithm	To consider different type of wastes	[100]
	PageRank algorithm + MILP	A model solved fast due to the use of the PageRank algorithm	Investigate more heuristic algorithms	[96]

Table 3. Optimization Models in SWM—the transportation stage.

Model Category	Model	Main Contribution	Limitations and Future Research Opportunities	Ref.
Exact Models	MILP	A model to find the best location for a transfer center with optimum cost and GHS emissions	Many assumptions were made to simplify the model; stochastic MSW rates can be used	[1]
	IP	A model to find the best location in terms of economics correlates with the place with the lowest CO ₂ emissions	Many assumptions were made to simplify the model to allow more complex cases to be considered	[59]
		A model to optimize routes to reduces the collection and transport costs	Use an efficient algorithm to solve larger-scale collection problems	[60]
	MILP	A model to find optimum fleet size and vehicle routes	Many assumptions were made to simplify the model to allow more complex cases to be considered	[52]
Approximate Models	FLP	A model to find optimum route for medical waste vehicles		[83]
Hybrid Models	MILP + modified Dijkstra's algorithm	A model to find the optimum route and optimum cost	Heuristic algorithms can be used to find a better solution	[102]
	GA+SCCP	A model to find optimal resources from separation to processing facilities	Environmental and social attributes of GHG emissions need to be considered in the model	[103]

Table 4. Optimization Models in SWM—the processing stage.

Model Category	Model	Main Contribution	Limitations and Future Research Opportunities	Ref.
Exact Models	LP	A model to optimize the processing protocol used and processing cost	Many assumptions were made to simplify the model to allow more complex cases to be considered	[4]
	IP and MILP	A model to find the optimal number of processing facilities	Many factors such as environmental and real-time waste generation rates could be considered	[32]
	LF	A model to find the optimal waste flow rate in processing facilities	Many assumptions were made to simplify the model to allow more complex cases to be considered	[41]
	MINLP	A model to link the GHG emissions to the waste treatment cost		[62]
	NLP	A model to find the optimum economic and environmental performance based on life cycle analysis	Many assumptions were made for the financial part that need to be improved	[64]
	MILP + MINLP	A model considering four scenarios between minimum cost and carbon emissions along with constant and variable capacity	Social impacts of different waste treatment technologies could be considered	[65]
	MINLP	A model to find optimum waste processing pathways	Must add the whole supply chain	[66]
Approximate Models	FLP	A model to locate the waste collection and processing facilities	Uncertainties in waste amount must be considered	[84]
	IFSQP	A model to optimally allocate waste for processing facilities	Environmental impact to be added as a constraint	[85]

The key findings in transportation within SWM processes focus on optimizing routes and locations to minimize both economic costs and CO₂ emissions (Table 3). Exact models, like IP, have highlighted a correlation between the economically optimal locations and the lowest CO₂ emissions, yet many assumptions simplify these models. Approaches to finding the optimal paths for reducing collection and transport costs call for more efficient algorithms to tackle larger-scale problems. Approximate models, such as FLP, have been applied to medical waste vehicles, identifying optimal routing under simplified assumptions. Hybrid models, like GA + SCCP, aim to optimize resource allocation from separation to

processing facilities, underscoring the need to integrate social and environmental considerations, particularly greenhouse gas emissions, into the modeling process. These studies point to significant opportunities for advancing SWM transportation models by incorporating more complex cases and considering broader environmental impacts.

The key findings from research on processing in SWM processes highlight the development of models to balance economic costs with environmental impacts, particularly greenhouse gas (GHG) emissions (Table 4). Models such as MINLP and NLP focus on linking GHG emissions to waste treatment costs and optimizing performance based on life cycle analysis, respectively, yet both acknowledge simplifications due to assumptions. MILP + MINLP models explore scenarios to balance costs and emissions, suggesting further investigation into the social impacts of waste treatment technologies. Additionally, models are proposed to optimize waste processing pathways and facility locations, with future research opportunities including the need to consider uncertainties in waste amounts, the entire supply chain, and the environmental impacts more comprehensively. These findings point towards an integrated approach that weighs economic, environmental, and social factors in SWM processing strategies.

The key findings in the recycling segment of SWM processes underscore the advancement of models designed to optimize the allocation of resources, materials, and logistics within recycling operations (Table 5). These models, ranging from dynamic models to various forms of mathematical programming such as MILP, NLP, and MINLP, facilitate the analysis of different materials, the usage of collection trucks, and the management of bin counts. They contribute significantly towards exploring the viability of a circular economy by considering economic, environmental, and social perspectives. Despite these advancements, a common limitation across these models is the reliance on simplifying assumptions to manage complexity, indicating a clear avenue for future research to incorporate more detailed and complex cases. This highlights an ongoing need to refine these models to better handle the intricate realities of recycling processes.

Table 5. Optimization Models in SWM—the recycling stage.

Model Category	Model	Main Contribution	Limitations and Future Research Opportunities	Ref.
Exact Models	MILP	A model that optimizes end-of-life costs and material recovery	More scenarios need to be investigated to help with generalizing the results	[22]
	Dynamic model	A model that enables the consideration of various materials, collection trucks, and bin counts	More complex cases need be considered	[70]
	MILP	A multi-level, multi-product (MILP) model	Many assumptions were made to simplify the model to allow more complex cases to be considered	[30,46,47]
	NLP	A model to find optimum resource allocation	To simplify the model, many assumptions were made to allow more complex cases to be considered	[63]
		A model to find maximum profit for recyclable waste		[104]
	MINLP	A model to explore the viability of the circular economy in managing MSW from economic, environmental, and social perspectives		[68]
	LFP	A flexible and naturally generic model	To simplify the model, many assumptions were made, to allow more complex cases to be considered	[40]

Table 6. Optimization models in SWM—all processes.

Model Category	Model	Main Contribution	Limitations and Future Research Opportunities	Ref.
Exact Models	LP	A model to optimize socioeconomic and environmental considerations	Many assumptions were used to simplify the model, so uncertainties in each stage must be considered	[29]
		A model to minimize the total cost of SWM		[105]
		A model to minimize the environmental impact of SWM		[106]
	MILP	A multi-objective model to optimize the cost and risk objectives	A stochastic model can be beneficial for the analysis	[37]
		A model to find maximum profits and minimize environmental effects of COVID-19 related waste		[43]
		A model to find optimal locations for the different facilities	Capacity variation in the different facilities needs to be considered	[44]
		A model that considers the most cost-effective approach involves strategically placing sites for collection and transfer centers at optimal locations	Uncertainty challenge of seasonal MSW supply	[45]
			Environmental and social aspects of GHG emissions need to be considered in the model	[51]
			The model considers different types of SW and collection vehicles	[53]
		A model that incorporates sustainability into the objective function to enhance recycling profits is considered	Extend the model to accommodate uncertainties by introducing stochastic	[97]
		A multi-objective model to the find maximum net profit while minimizing environmental effects	Different scenarios need to be considered	[54]
		A multi-objective model to optimize the balance between the cost and environmental impacts of GHG emissions	Consider long-term decisions and multi-periods	[48]
	MINLP	A model to find the minimum costs of waste collection, recycling, and disposal facilities	Many assumptions were made to simplify the model to allow uncertainties of all stages to be considered	[31]
	Dynamic MILP	A dynamic model that considers SWMSs' cost dependance in terms of time	Uncertainties of waste generation need to be considered	[71]
Approximate Models	SP	A model to find the optimum GHG emissions and social cost of a carbon footprint	Many assumptions were made to simplify the model to allow all types of waste to be considered	[3]
	FLP	A unique multi-objective model to optimize the economic–environment–energy in SWMSs	Many assumptions were made to simplify the model to allow more complex cases to be considered	[38,82,86]
	SECCP	A stochastic model that considers bi-random variables that enriched the solution	Need to tackle more complex scenarios	[75]
	FSLP	A model dealing with the interactions between dual levels and high uncertainty	Promising as it can provide higher convenience to decision-makers than the existing models	[80]
	SMILP	A model to find the optimum annual cost, optimum material distribution, optimal waste treatment technology, and optimal capacity of treatment facilities	Considers multi-period MILP models	[76]
	SFQP	A model allowed dual uncertainty to be represented as probability	The generated method can be used for other types of resource management	[81]
	SP	Stochastic and deterministic models were compared	Consider multistage stochastic formulation with parameters uncertainties	[78,79]
Hybrid Models	MILP+ MOSA-MOIWOA algorithm	A model confirmed that the performance of the MOSA-MOIWOA algorithm outperformed classical metaheuristics	Need to tackle more complex scenarios	[94]

Across the spectrum of SWM processes, a significant focus has been placed on developing models to enhance efficiency, sustainability, and profitability (Table 6). The MILP models stand out for their contributions towards maximizing profits and minimizing environmental impacts, including those related to COVID-19 waste. Challenges include the need for stochastic models to better address uncertainties in waste volume and facility capacity. Furthermore, multi-objective models aim to balance cost, greenhouse gas emissions, and environmental impact, suggesting a move towards more dynamic and adaptive strategies. Approximate and hybrid models, employing novel approaches and algorithms like MOSA-MOIWOA, show promise in refining SWM by considering economic, environmental, and energy factors. The research collectively indicates a pressing need to embrace complex scenarios, long-term planning, and the integration of uncertainties into SWM modeling to drive future innovations.

4. Conclusions, Recommendations, and Future Research Directions

The systematic review conducted on optimization models and techniques in SWM has provided valuable perceptions into the state of the field and its potential for improving the efficiency and sustainability of waste management systems. The review identified a wide range of optimization approaches employed in different phases of waste management, including waste collection and transportation, waste treatment and disposal, and resource recovery. Also, there is a lack of research on SWM within the Gulf region despite the rapid growth and urbanization experienced by countries in this region. Serious efforts must be made to focus on the topic due to the importance of SWM in ensuring environmental sustainability and public health.

Overall, the findings suggest that optimization techniques have the potential to significantly enhance SWM practices by optimizing routes, schedules, and resource allocation, thereby reducing costs, minimizing environmental impacts, and improving overall system performance. The review revealed that mathematical modeling, operations research, and artificial intelligence are among the most utilized tools in SWM optimization.

The practical implications of this work can be understood in several key areas. This study highlights significant advances in optimizing waste collection routes and schedules, leading to reduced fuel consumption, lower emissions, and cost savings for municipalities. Optimizing the placement of waste bins minimizes the distance and time required for collection, enhancing overall efficiency and service quality. In terms of waste treatment and disposal, optimization techniques help to select appropriate landfill sites while considering environmental impact, transportation costs, and local regulations, as well as optimizing waste-to-energy conversion methods, providing sustainable energy sources, and reducing reliance on fossil fuels. This study focuses on improving sorting and recycling operations, boosting valuable material recovery rates, minimizing landfill trash, and promoting circular economy concepts, all of which increase sustainability.

Moreover, implementing optimization techniques may result in considerable cost savings across all parts of MSWM system, while also assisting with improved financial planning and budgeting. Environmentally, optimizing waste management operations minimizes the total environmental impact, resulting in fewer greenhouse gas emissions and pollution, while also increasing public health and safety by minimizing hazardous waste exposure and promoting cleanliness. This study offers useful insights for politicians and waste management experts, enabling data-driven decisions and personalized responses to unique geographical needs.

Despite the promising outcomes observed, this review also detected several challenges and constraints involved in the application of optimization techniques in solid waste management. These include data availability and quality, stakeholder involvement, technological constraints, and the need for context-specific solutions. Focusing on these challenges will be crucial for the effective implementation of optimization techniques in real-world waste management scenarios. Therefore, use of IoT for data collection and categorization is recommended to improve the quality of collected data and the formulation of the problem

as it can consider the heterogeneity of all solid waste flows. Also, greater use of dynamic models must be considered in future research as it helps with assessing different scenarios and strategies while considering the time dependency of variables, which guarantees more realistic formulation of the problem. Moreover, one of the interesting realizations of this systematic review is that there is limited focus on the waste generation stage, even though it is the most significant stage in the whole process, and studying optimization techniques that can be utilized to find optimum solutions to minimize waste generation such as greater introduction of IoT that can classify the waste to different types can help researchers to find patterns that can be helpful in the optimization process.

In conclusion, the systematic review highlights the significant potential of optimization models and techniques in improving solid waste management systems. The findings enhance the current knowledge base by offering an overview of the recent developments in the field, identifying research gaps, and suggesting future directions. The review underscores the importance of interdisciplinary collaborations, evidence-based decision-making, and the adoption of tailored approaches to address the complex challenges associated with waste management. Policymakers, waste management practitioners, and researchers can benefit from the insights provided in this systematic review to inform the design and execution of sustainable strategies for waste management. Further research and innovation in optimization techniques, coupled with effective stakeholder engagement, will be crucial for advancing the field and achieving more efficient, cost-effective, and environmentally friendly waste management systems in the future.

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